

Author(s):sarah

Published on:07-Jul-2026

Requirement Set: StakeholderNeeds	3
1 Soup menu	4
2 Small crew.....	4
3 Galactic shipping.....	5
4 Food safety	5
5 Transit-durable packaging.....	6
6 Inventory tracking	6
7 Facility budgets	7
8 Gravity range.....	7
9 Delivery fleet support.....	8
10 Ingredient Storage	9
11 Cooking infrastructure.....	9
12 Internal material transport.....	10
13 Operational lifecycle.....	10
14 Fault tolerance	11
15 System coordination	12
Requirement Set: SystemRequirements.....	13
1 Recipe count.....	14
2 Throughput.....	14
3 Automation level.....	15
4 Operator count	16
5 Shipping manifest.....	16
6 Transport loading time	17
7 Contamination detection	17
8 Serving temperature	18
9 Container seal life	19
10 Inventory accuracy	20
11 Mass budget.....	20
12 Power budget.....	21
13 Cost budget	22
14 Volume budget.....	23
15 Gravity operating range.....	23
16 Structural 12 g tolerance	24
17 Concurrent rocket support.....	25
18 Rocket turnaround	26
19 Rocket refueling	27
20 Temperature-controlled storage	27
21 Ingredient storage capacity	28
22 Cooking capacity.....	29
23 Internal transfer time.....	30

24 Internal transport automation.....	30
25 Startup readiness.....	31
26 Fault-induced throughput degradation.....	31
27 Production coordination	32
28 Preparation zone	33
Artifact List.....	34

Requirement Set: StakeholderNeeds

Description

Attributes

Filepath	D:\dev\galactic-soup\GalacticSoup\requirements\StakeholderNeeds.sreqx
Revision	1
Created by	sarah
Created on	05-Jul-2026 12:23:21
Modified by	sarah
Modified on	05-Jul-2026 12:23:30

Implementation Status

Total	Implemented	Justified	None
15	0	0	15

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
15	0	0	0	0	15

Change Information No change issue detected.

1 Soup menu

Requirement Type Functional

ID SN-GS-001

Description

Kitchen shall produce a menu of distinct soup varieties selectable by operators.

Rationale

Customers expect variety across the menu.

Revision Information

SID	1	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)

 [SR-GS-001 Recipe count](#) (⇒Derives)

 [SR-GS-002 Throughput](#) (⇒Derives)

 [SR-GS-028 Preparation zone](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

2 Small crew

Requirement Type Functional

ID SN-GS-002

Description

Kitchen shall be operable by the small crew of beings on site.

Rationale

Only 5 beings staff the facility.

Revision Information

SID	2	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)

 [SR-GS-003 Automation level](#) (⇒Derives)

 [SR-GS-004 Operator count](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

3 Galactic shipping

Requirement Type Functional

ID SN-GS-003

Description

Kitchen shall dispatch finished soup to customers across the galaxy.

Rationale

Customers are distributed across many worlds.

Revision Information

SID	3	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)

 [SR-GS-005 Shipping manifest](#) (⇒Derives)

 [SR-GS-006 Transport loading time](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

4 Food safety

Requirement Type Functional

ID SN-GS-004

Description

Kitchen shall deliver safe, uncontaminated soup at correct serving temperature.

Rationale

Safety and quality are non-negotiable.

Revision Information

SID	4	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)

 [SR-GS-007 Contamination detection](#) (⇒Derives)

 [SR-GS-008 Serving temperature](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

5 Transit-durable packaging

Requirement Type Functional

ID SN-GS-005

Description

Packaging shall keep soup intact during long-distance interstellar transit.

Rationale

Shipments take weeks to reach far customers.

Revision Information

SID	5	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)

 [SR-GS-009 Container seal life](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

6 Inventory tracking

Requirement Type Functional

ID SN-GS-006

Description

Kitchen shall track ingredient stock so the crew knows what to reorder.

Rationale

Stock-outs halt production; crew needs visibility.

Revision Information

SID	6	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)

 [SR-GS-010 Inventory accuracy](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

7 Facility budgets

Requirement Type Functional

ID SN-GS-007

Description

Kitchen shall fit within facility mass, volume, power, and cost budgets.

Rationale

Deployment site imposes hard resource limits.

Revision Information

SID	7	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)

 [SR-GS-011 Mass budget](#) (⇒Derives)

 [SR-GS-012 Power budget](#) (⇒Derives)

 [SR-GS-013 Cost budget](#) (⇒Derives)

 [SR-GS-014 Volume budget](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

8 Gravity range

Requirement Type Functional

ID SN-GS-008

Description

Kitchen shall operate across the range of gravitational environments where customer worlds are located.

Rationale

Facility may be deployed on any of a diverse set of worlds.

Revision Information

SID	8	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.sreqx](#)

 [SR-GS-015 Gravity operating range](#) (⇒Derives)

 [SR-GS-016 Structural 12 g tolerance](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

9 Delivery fleet support

Requirement Type Functional

ID SN-GS-009

Description

Kitchen shall operate and include a fleet of delivery rockets for transporting finished soup to customers and receiving inbound deliveries.

Rationale

Dedicated infrastructure required for frequent launch, landing, loading, and refueling operations.

Revision Information

SID	9	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.sreqx](#)

 [SR-GS-017 Concurrent rocket support](#) (⇒Derives)

 [SR-GS-018 Rocket turnaround](#) (⇒Derives)

 [SR-GS-019 Rocket refueling](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

10 Ingredient Storage

Requirement Type Functional

ID SN-GS-010

Description

Kitchen shall store a variety of delivered ingredients under appropriate temperature and handling conditions.

Rationale

Need safe, flexible storage of both perishable and shelf-stable ingredients.

Revision Information

SID	10	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.sreqx](#)

 [SR-GS-020 Temperature-controlled storage](#) (⇒Derives)

 [SR-GS-021 Ingredient storage capacity](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

11 Cooking infrastructure

Requirement Type Functional

ID SN-GS-011

Description

Kitchen shall include cooking infrastructure capable of producing soup at required production rates.

Rationale

Throughput targets cannot be met without sufficient cooking capacity matched to demand.

Revision Information

SID	11	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)
 [SR-GS-022 Cooking capacity](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

12 Internal material transport

Requirement Type Functional

ID SN-GS-012

Description

Kitchen shall transport ingredients and products internally between receiving, storage, cooking, packaging, and delivery areas.

Rationale

Required to meet throughput targets.

Revision Information

SID	12	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)
 [SR-GS-023 Internal transfer time](#) (⇒Derives)
 [SR-GS-024 Internal transport automation](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

13 Operational lifecycle

Requirement Type Functional

ID SN-GS-013

Description

Kitchen shall support startup, steady operation, and shutdown across its deployment lifecycle.

Rationale

System must operate safely outside of nominal steady-state conditions.

Revision Information

SID	13	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)
 [SR-GS-025 Startup readiness](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

14 Fault tolerance

Requirement Type Functional

ID SN-GS-014

Description

Kitchen shall continue operating or degrade safely in the presence of faults.

Rationale

Production continuity and safety must be maintained with minimal crew intervention.

Revision Information

SID	14	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.slreqx](#)
 [SR-GS-026 Fault-induced throughput degradation](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

15 System coordination

Requirement Type Functional

ID SN-GS-015

Description

Kitchen shall coordinate cooking, packaging, storage, and transport activities to meet operational goals.

Rationale

High automation requires integrated system-level control.

Revision Information

SID	15	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:22
Modified by	sarah	Modified on	05-Jul-2026 12:23:22

Change Information No change issue detected.

Links

Artifact [SystemRequirements.sreqx](#)

 [SR-GS-027 Production coordination](#) (⇒Derives)

Implementation Status

Total	Implemented	Justified	None
1	0	0	1

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

Requirement Set: SystemRequirements

Description

Attributes

Filepath	D:\dev\galactic-soup\GalacticSoup\requirements\SystemRequirements.slreqx
Revision	1
Created by	sarah
Created on	05-Jul-2026 12:23:23
Modified by	sarah
Modified on	07-Jul-2026 16:17:15

Implementation Status

Total	Implemented	Justified	None
28	28	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
28	2	0	0	0	26

Change Information No change issue detected.

1 Recipe count

Requirement Type Functional

ID SR-GS-001

Description

System shall cook at least 8 distinct recipes selectable at runtime.

Rationale

Eight recipes gives menu variety while staying within ingredient storage and recipe-validation budgets.

Revision Information

SID	1	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:5 (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-001 Soup menu](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

2 Throughput

Requirement Type Functional

ID SR-GS-002

Description

System shall sustain total throughput of at least 200 bowls/hour.

Rationale

200 bowls/hour matches expected peak order rate aggregated across customer worlds.

Revision Information

SID	2	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate_2.slreqx:7 (↔Refined by)

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:5 (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-001 Soup menu](#) (↔Derived from)

Artifact [GalacticSoupSystemTests.mldatx](#)

📄 [HyperCook nominal](#) (↔Verified by)

📄 [EverSimmer nominal](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

3 Automation level

Requirement Type Functional

ID SR-GS-003

Description

System shall achieve average automation level of at least 0.8 across components.

Rationale

0.8 automation lets the 5-operator crew cover a 24/7 facility with rotation and breaks.

Revision Information

SID	3	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate_2.slreqx:8 (↔Refined by)

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:11 (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-002 Small crew](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
-------	--------	-----------	--------	------------	------

1	0	0	0	0	1
---	---	---	---	---	---

4 Operator count

Requirement Type Functional

ID SR-GS-004

Description

System shall require at most 5 concurrent operators at peak load.

Rationale

Facility berthing and life support are sized for a 5-person crew.

Revision Information

SID	4	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate_2.slreqx:9 (←Refined by)

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:11 (←Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-002 Small crew](#) (←Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

5 Shipping manifest

Requirement Type Functional

ID SR-GS-005

Description

System shall generate a shipping manifest per batch including destination.

Rationale

Manifest is required for customs, transport routing, and recipient confirmation on delivery.

Revision Information

SID	5	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:8 (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-003 Galactic shipping](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

6 Transport loading time

Requirement Type Functional

ID SR-GS-006

Description

System shall load packaged soup onto transport within 10 minutes of packaging.

Rationale

10-minute window keeps loading pipeline aligned with the 200 bowls/hour production rate.

Revision Information

SID	6	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:8 (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-003 Galactic shipping](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

7 Contamination detection

Requirement Type Functional

ID SR-GS-007

Description

System shall detect contamination before sealing with at least 99% sensitivity.

Rationale

99% sensitivity balances false-positive rework cost against the downstream cost of a contaminated shipment.

Revision Information

SID	7	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:6 (↔Implemented by)

Artifact [PhysicalEverSimmer.slx](#)

 [CellQCSensor](#) (↔Implemented by)

 [CellQCSensor](#) (↔Implemented by)

 [CellQCSensor](#) (↔Implemented by)

Artifact [PhysicalHyperCook.slx](#)

 [InlineQCScanner](#) (↔Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

 [QCBench](#) (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

 [SN-GS-004 Food safety](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

8 Serving temperature

Requirement Type Functional

ID SR-GS-008

Description

System shall verify soup temperature in 70-95 C before QC sign-off.

Rationale

70-95 C is the safe serving range: above pathogen-kill threshold and below scald risk.

Revision Information

SID	8	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23

Modified by	sarah	Modified on	05-Jul-2026 12:23:23
--------------------	-------	--------------------	----------------------

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:6 (↔Implemented by)

Artifact [PhysicalEverSimmer.slx](#)

☐ [CellQCSensor](#) (↔Implemented by)

☐ [CellQCSensor](#) (↔Implemented by)

☐ [CellQCSensor](#) (↔Implemented by)

Artifact [PhysicalHyperCook.slx](#)

☐ [InlineQCScanner](#) (↔Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

☐ [QCBench](#) (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

☒ [SN-GS-004 Food safety](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

9 Container seal life

Requirement Type Functional

ID SR-GS-009

Description

System shall seal containers rated for 30-day interstellar transit.

Rationale

30 days covers typical transit time to the farthest customer worlds with margin.

Revision Information

SID	9	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:7 (↔Implemented by)

Artifact [PhysicalEverSimmer.slx](#)

☐ [CellPackager](#) (↔Implemented by)

☐ [CellPackager](#) (↔Implemented by)

 [CellPackager](#) (↔Implemented by)

Artifact [PhysicalHyperCook.slx](#)

 [HighSpeedPackagingLine](#) (↔Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

 [SemiAutoPackager](#) (↔Implemented by)

Artifact [StakeholderNeeds.slsreqx](#)

 [SN-GS-005 Transit-durable packaging](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

10 Inventory accuracy

Requirement Type Functional

ID SR-GS-010

Description

System shall track ingredient inventory with at most 1% stock error.

Rationale

1% stock error keeps reorder decisions accurate enough to avoid production-halting stock-outs.

Revision Information

SID	10	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:3 (↔Implemented by)

Artifact [StakeholderNeeds.slsreqx](#)

 [SN-GS-006 Inventory tracking](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

11 Mass budget

Requirement Type Functional

ID SR-GS-011

Description

Total system mass shall not exceed 15000 kg.

Rationale

Allocated from the facility deployment mass budget.

Revision Information

SID	11	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupComplianceGate.slx](#)

 GalacticSoupComplianceGate_2.slreqx:3 (↔Refined by)

Artifact [PhysicalEverSimmer.slx](#)

 [PhysicalEverSimmer](#) (↔Implemented by)

Artifact [PhysicalHyperCook.slx](#)

 [PhysicalHyperCook](#) (↔Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

 [PhysicalLeanBroth](#) (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

 [SN-GS-007 Facility budgets](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

12 Power budget

Requirement Type Functional

ID SR-GS-012

Description

Total system power draw shall not exceed 500 kW.

Rationale

Allocated from the facility power budget.

Revision Information

SID	12	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate_2.slreqx:4 (⇐Refined by)

Artifact [PhysicalEverSimmer.slx](#)

☐ [PhysicalEverSimmer](#) (⇐Implemented by)

Artifact [PhysicalHyperCook.slx](#)

☐ [PhysicalHyperCook](#) (⇐Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

☐ [PhysicalLeanBroth](#) (⇐Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

☐ [SN-GS-007 Facility budgets](#) (⇐Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

13 Cost budget

Requirement Type Functional

ID SR-GS-013

Description

Total system cost shall not exceed 2000000 credits.

Rationale

Allocated from the facility capital cost budget.

Revision Information

SID	13	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate_2.slreqx:5 (⇐Refined by)

Artifact [PhysicalEverSimmer.slx](#)

☐ [PhysicalEverSimmer](#) (⇐Implemented by)

Artifact [PhysicalHyperCook.slx](#)

☐ [PhysicalHyperCook](#) (⇐Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

☐ [PhysicalLeanBroth](#) (⇐Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

 [SN-GS-007 Facility budgets](#) ( Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

14 Volume budget

Requirement Type Functional

ID SR-GS-014

Description

Total system volume shall not exceed 400 m³.

Rationale

Allocated from the facility floor-space and volume budget.

Revision Information

SID	14	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupComplianceGate.slx](#)

 GalacticSoupComplianceGate_2.slreqx:6 ( Refined by)

Artifact [PhysicalEverSimmer.slx](#)

 [PhysicalEverSimmer](#) ( Implemented by)

Artifact [PhysicalHyperCook.slx](#)

 [PhysicalHyperCook](#) ( Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

 [PhysicalLeanBroth](#) ( Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

 [SN-GS-007 Facility budgets](#) ( Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

15 Gravity operating range

Requirement Type Functional

ID SR-GS-015

Description

System shall perform all cooking, packaging, and shipping functions nominally across ambient gravity 0.1 g to 12 g.

Rationale

0.1-12 g spans the full ambient gravity range across candidate customer worlds.

Revision Information

SID	15	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupComplianceGate.slx](#)

⚠ GalacticSoupComplianceGate_2.slreqx:10 (↔Refined by)

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:12 (↔Implemented by)

Artifact [PhysicalEverSimmer.slx](#)

☐ [GravityCompMesh](#) (↔Implemented by)

Artifact [PhysicalHyperCook.slx](#)

☐ [GravityCompensatorArray](#) (↔Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

☐ [GravityCompUnit](#) (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-008 Gravity range](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

16 Structural 12 g tolerance

Requirement Type Functional

ID SR-GS-016

Description

System structure and mounts shall withstand sustained 12 g loading without permanent deformation.

Rationale

12 g is the worst-case sustained ambient gravity among customer worlds.

Revision Information

SID	16	Revision	1
------------	----	-----------------	---

Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

- Artifact** [GalacticSoupFunctional.slx](#)
 GalacticSoupFunctional:12 (↔Implemented by)
Artifact [PhysicalEverSimmer.slx](#)
☐ [GravityCompMesh](#) (↔Implemented by)
Artifact [PhysicalHyperCook.slx](#)
☐ [GravityCompensatorArray](#) (↔Implemented by)
Artifact [PhysicalLeanBroth.slx](#)
☐ [GravityCompUnit](#) (↔Implemented by)
Artifact [StakeholderNeeds.slsreqx](#)
 [SN-GS-008 Gravity range](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

17 Concurrent rocket support

Requirement Type Functional

ID SR-GS-017

Description

System shall support simultaneous handling of at least 3 delivery rockets for loading or unloading.

Rationale

Parallel handling prevents outbound shipping from blocking inbound ingredient deliveries.

Revision Information

SID	17	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

- Artifact** [GalacticSoupFunctional.slx](#)
 GalacticSoupFunctional:9 (↔Implemented by)
Artifact [PhysicalEverSimmer.slx](#)
☐ [TriplePadPort](#) (↔Implemented by)

Artifact [PhysicalHyperCook.slx](#)

☐ [LaunchPadComplex](#) (↔Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

☐ [TriPadLandingField](#) (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

☐ [SN-GS-009 Delivery fleet support](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

18 Rocket turnaround

Requirement Type Functional

ID SR-GS-018

Description

System shall complete loading or unloading of a delivery rocket within 20 minutes.

Rationale

Fast turnaround maximizes fleet utilization and maintains alignment with production throughput.

Revision Information

SID	18	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:9 (↔Implemented by)

Artifact [PhysicalEverSimmer.slx](#)

☐ [AutoCargoLoader](#) (↔Implemented by)

Artifact [PhysicalHyperCook.slx](#)

☐ [CargoLoaderGantry](#) (↔Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

☐ [SharedCraneDock](#) (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

☐ [SN-GS-009 Delivery fleet support](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

19 Rocket refueling

Requirement Type Functional**ID** SR-GS-019**Description**

System shall provide on-site refueling capability for supported delivery rockets.

Rationale

On-site refueling minimizes logistics dependencies and enables rapid redeployment.

Revision Information

SID	19	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.**Links****Artifact** [GalacticSoupFunctional.slx](#) GalacticSoupFunctional:9 (↔Implemented by)**Artifact** [PhysicalEverSimmer.slx](#) [AutoRefuelCell](#) (↔Implemented by)**Artifact** [PhysicalHyperCook.slx](#) [RefuelingStation](#) (↔Implemented by)**Artifact** [PhysicalLeanBroth.slx](#) [RefuelSkid](#) (↔Implemented by)**Artifact** [StakeholderNeeds.slreqx](#) [SN-GS-009 Delivery fleet support](#) (↔Derived from)**Implementation Status**

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

20 Temperature-controlled storage

Requirement Type Functional**ID** SR-GS-020**Description**

System shall provide separate storage zones for cold and room-temperature ingredients.

Rationale

Different ingredient classes require different storage conditions to maintain safety and quality.

Revision Information

SID	20	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:2 (↔Implemented by)

Artifact [PhysicalEverSimmer.slx](#)

☐ [DualZoneStore](#) (↔Implemented by)

Artifact [PhysicalHyperCook.slx](#)

☐ [ColdStorageVault](#) (↔Implemented by)

Artifact [PhysicalLeanBroth.slx](#)

☐ [ColdStoreLocker](#) (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-010 Ingredient Storage](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

21 Ingredient storage capacity

Requirement Type Functional

ID SR-GS-021

Description

System shall store at least 72 hours of ingredients at nominal production rate.

Rationale

72 hours of buffer mitigates delivery delays without exceeding storage mass and volume budgets.

Revision Information

SID	21	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)
 GalacticSoupFunctional:2 (↔Implemented by)
Artifact [PhysicalEverSimmer.slx](#)
☐ [DualZoneStore](#) (↔Implemented by)
Artifact [PhysicalHyperCook.slx](#)
☐ [AmbientStorageSilo](#) (↔Implemented by)
Artifact [PhysicalLeanBroth.slx](#)
☐ [DryGoodsRack](#) (↔Implemented by)
Artifact [StakeholderNeeds.slsreqx](#)
☐ [SN-GS-010 Ingredient Storage](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

22 Cooking capacity

Requirement Type Functional

ID SR-GS-022

Description

System shall cook soup at a sustained rate sufficient to meet overall throughput requirements.

Rationale

Cooking must not be the bottleneck in achieving 200 bowls/hour system throughput.

Revision Information

SID	22	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)
 GalacticSoupFunctional:5 (↔Implemented by)
Artifact [StakeholderNeeds.slsreqx](#)
☐ [SN-GS-011 Cooking infrastructure](#) (↔Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

23 Internal transfer time

Requirement Type Functional

ID SR-GS-023

Description

System shall transport ingredients from storage to cooking at a rate sufficient to sustain nominal cooking throughput.

Rationale

Cooking throughput must not be limited by internal ingredient transport.

Revision Information

SID	23	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:10 (⇐Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

 [SN-GS-012 Internal material transport](#) (⇐Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

24 Internal transport automation

Requirement Type Functional

ID SR-GS-024

Description

Internal material transport shall be automated for at least 90% of transfers.

Rationale

High automation supports low crew size and reduces operator workload.

Revision Information

SID	24	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:10 (⇐Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-012 Internal material transport](#) (⇐Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

25 Startup readiness

Requirement Type Functional

ID SR-GS-025

Description

System shall reach nominal operating throughput within a defined startup period after activation.

Rationale

Ensures the system can transition from inactive to productive operation without extended downtime.

Revision Information

SID	25	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:11 (⇐Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

📄 [SN-GS-013 Operational lifecycle](#) (⇐Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

26 Fault-induced throughput degradation

Requirement Type Functional

ID SR-GS-026

Description

System shall prevent uncontrolled termination of production due to single internal component faults.

Rationale

Should tolerate individual component failures without complete loss of production.

Revision Information

SID	26	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:11 (↔Implemented by)

Artifact [StakeholderNeeds.slsreqx](#)

📄 [SN-GS-014 Fault tolerance](#) (↔Derived from)

Artifact [GalacticSoupSystemTests.mldatx](#)

📄 [EverSimmer worst fault](#) (↔Verified by)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	1	0	0	0	0

27 Production coordination

Requirement Type Functional

ID SR-GS-027

Description

System shall coordinate internal operations to satisfy throughput, safety, and logistical constraints concurrently.

Rationale

Ensures that throughput, safety, and logistics objectives are met simultaneously in a highly automated system.

Revision Information

SID	27	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

⚠ GalacticSoupFunctional:11 (↔Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

 [SN-GS-015 System coordination](#) (←Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

28 Preparation zone

Requirement Type Functional

ID SR-GS-028

Description

System shall include a separate ingredient preparation zone supporting chopping and weighing of ingredients, physically isolated from storage and cooking zones. The zone shall sustain prep throughput of at least 200 bowls/hour equivalent.

Rationale

A dedicated prep zone keeps raw/prepped materials separated and sized so it does not bottleneck the cooking line.

Revision Information

SID	28	Revision	1
Created by	sarah	Created on	05-Jul-2026 12:23:23
Modified by	sarah	Modified on	05-Jul-2026 12:23:23

Change Information No change issue detected.

Links

Artifact [GalacticSoupFunctional.slx](#)

 GalacticSoupFunctional:4 (←Implemented by)

Artifact [StakeholderNeeds.slreqx](#)

 [SN-GS-001 Soup menu](#) (←Derived from)

Implementation Status

Total	Implemented	Justified	None
1	1	0	0

Verification Status

Total	Passed	Justified	Failed	Unexecuted	None
1	0	0	0	0	1

Appendix

Artifact List

Requirement Set files:

#	Name	Folder	Revision
1	SystemRequirements.slreqx	D:\dev\galactic-soup\GalacticSoup\requirements	1
2	StakeholderNeeds.slrreqx	D:\dev\galactic-soup\GalacticSoup\requirements	1

Simulink models:

#	Name	Folder	Version
1	GalacticSoupFunctional.slx	D:\dev\galactic-soup\GalacticSoup\architecture	Unloaded.
2	GalacticSoupComplianceGate.slx	D:\dev\galactic-soup\GalacticSoup\architecture	Unloaded.
3	PhysicalEverSimmer.slx	D:\dev\galactic-soup\GalacticSoup\architecture	1.12
4	PhysicalHyperCook.slx	D:\dev\galactic-soup\GalacticSoup\architecture	1.10
5	PhysicalLeanBroth.slx	D:\dev\galactic-soup\GalacticSoup\architecture	1.10

Simulink Test files:

#	Name	Folder	File timestamp
1	GalacticSoupSystemTests.mldatx	D:\dev\galactic-soup\GalacticSoup\tests	07-Jul-2026 15:49:07